

Data processing

Notebook inputs
Raw EEG, event tables,
analysis-ready trial labels

Stimulus-locked preprocessing
band-pass 1–30 Hz
resample to 128 Hz
2 s epochs per trial

Epoch cache
trial × channel × time tensor
labels, groups, trial index

Feature engineering and model selection

Grouped model selection
StratifiedGroupKFold
participant-aware folds

Feature engineering
CSP spatial filters
log-variance component features

Hyperparameter sweep
compare 2, 4, 6, 8, 10, 12
CSP components

Modeling and evaluation

Classifier
LDA with shrinkage

Grouped evaluation
dummy baseline vs CSP+LDA
ROC AUC, accuracy, F1

Generalization evaluation
Leave-one-subject-out
subject-wise held-out testing

Reference check
within-subject CV
for separability comparison